



One of the most complete of the original Roman bridges, the Pons Aelius was built by Emperor Hadrian to provide a processional route to his mausoleum (now the Castel Sant'Angelo). In its original form the bridge had eight arches.



Ptolemy's map

The coordinates and topographic lists in Ptolemy's *Almagest* enabled maps to be composed of his view of the world. This map dates from 1492.

are on mathematical geography and astronomy. In *Geography*, he gave a description of the known world, including **coordinates for longitude and latitude** (the latter derived from the length of the longest day) and gave instructions for the creation of a world map. In *Mathematical Compendium*, also known as *Almagest*, Ptolemy presented a star catalog with over 1,000 listed stars and 48 constellations. He refined the **theory of the celestial spheres**, introducing additional epicycles to explain irregularities in the motion of the Sun and the Moon and the apparent retrograde motion of certain planets, when they appear to orbit in a contrary direction to other bodies in the Solar System. He was the first astronomer to **convert observational data into a mathematical model** to back up his theories, using spherical trigonometry to do so. His model of the Solar System remained the basis of astronomical theory until the Renaissance.

In 169, **Claudius Galen** became personal physician to the Roman emperor Marcus Aurelius. Galen specialized in anatomy and had earlier worked as a surgeon to

of the weight-supporting arch, and used it in bridge-building. In around 104, the engineer Apollodorus of Damascus had **constructed a great bridge across the Danube** to facilitate the Emperor Trajan's invasion of Dacia (modern Romania).

Trajan's bridge was destroyed in c.120 by his successor, Hadrian, who himself had several great bridges built, including the Pons Aelius in Rome in c.134.

The best-known works of Greek-Roman astronomer **Ptolemy of Alexandria** (c.90–168)

CLAUDIUS GALEN (c.130–c.210)

Born in the ancient Greek city of Pergamum, Claudius Galen consolidated the works of his predecessors to create a single scientific framework. His insistence on direct observation of the body cut across his view that each of the body's organs functioned according to a divinely ordained scheme. He wrote 350 medical works.



a gladiatorial school, where he gained valuable knowledge of human physiology and surgery. He championed the theory that the **body had four basic humors** (see panel, below).

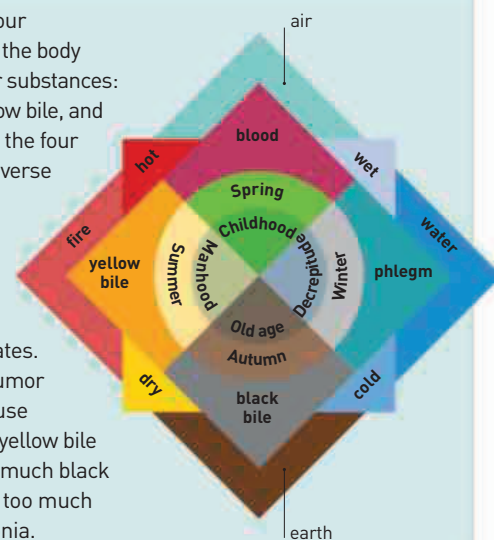
The Bakshali manuscript, found in what is now Pakistan, dates from around 200 and contains instructions for the computation of square roots. It is probably the **earliest document to use a specific sign for zero in the decimal system**, making it the first complete decimal notation with a single sign for each number value. This system spread westward

through the Arabic world, acquiring the popular name of Arabic numerals.

Chinese mathematics had made significant advances by the time *Jiuzhang Suanshu* (*Nine Chapters on the Mathematical Art*) was in existence in 179. It included rules for **calculating the area of arcs of circles** and the volume of solid figures such as cones, and for the treatment of vulgar fractions (written in the form x/y). It contained instructions for the **calculation of linear equations**, including the earliest appearance of equations with negative numbers.

THE FOUR HUMORS

The theory of the four humors stated that the body is composed of four substances: blood, phlegm, yellow bile, and black bile. In blood, the four elements of the universe (fire, air, earth, and water) are mixed equally, while in the other humors, one element predominates. An excess of one humor was believed to cause disease. Too much yellow bile led to jaundice, too much black bile to leprosy, and too much phlegm to pneumonia.



105 Traditional date of invention of paper in China by Cai Lun

c.134 Construction of the Pons Aelius bridge is completed

127–141 Ptolemy makes astronomical observations in Alexandria

169 Galen becomes personal physician to Emperor Marcus Aurelius

c.200 Bakshali manuscript contains first specific sign for zero

c.200 Distillation by freezing is invented by steppe nomads

c.200 Heavy cavalry armor is developed in Parthia

After 200 Knitting is invented in Syria